SUMANTH R

Chennai, Tamil Nadu
B.Tech. in Computer Science and Engineering (AIML)
SRM Institute of Science and Technology

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Education

Degree	Institute	CGPA	Duration
B.Tech., Computer Science and Engineering (AIML)	SRM Institute of Science and Technology, Chennai, Tamil Nadu	8.98 / 10	*2023 – 2027*

Experience

- **Works Applications India Pvt. Ltd.** *Dec 2025 – Jan 2026*
  Software Engineering Intern On-site
  - Built a full-stack application using **Java, Spring Boot, MySQL, and React**, developing 5+ REST APIs for business workflows.
  - Designed and optimized **SQL queries, database workflows, and indexing**, reducing backend response time by **25%** during testing.
  - Integrated backend services with frontend components, implementing **API validation, error handling, and database operations** across application modules.

- **Hyphiin** *Apr 2026 – Jun 2026*
  Software Development Engineering Intern Remote
  - Developed **5+ REST APIs** for application workflows, integrating authentication, validation, business logic, and database operations.
  - Contributed to **3+ full-stack modules**, working across backend services, frontend components, and database-driven workflows.
  - Resolved **15+ bugs and integration issues** through systematic debugging and testing, improving application reliability.

Projects

- **Distributed Job Scheduler** *2026*
  Technologies: Python, FastAPI, PostgreSQL, Redis, Docker, AWS, Linux
  - Developed a distributed scheduling system with **REST APIs, task queues, worker coordination, retries, and failure handling**.
  - Implemented persistent job tracking, concurrency control, retry mechanisms, and **idempotent execution** using PostgreSQL and Redis.
  - Added structured logging and error-handling workflows to support **failure diagnosis, troubleshooting, and recurring error analysis**.

- **Network Troubleshooting & Monitoring Toolkit** *2026*
  Technologies: Python, Linux, TCP/IP, DNS, HTTP/HTTPS, Socket Programming
  - Developed Python utilities to perform **DNS resolution, TCP connectivity checks, HTTP endpoint testing, and network diagnostics** on Linux.
  - Implemented structured diagnostic output to identify connectivity, timeout, DNS, and service availability issues.
  - Added logging and error classification to simplify **root-cause analysis and technical troubleshooting**.

- **API Integration & Troubleshooting Platform** *2025*
  Technologies: Java, Spring Boot, REST APIs, MySQL, Docker, Linux
  - Built REST APIs with authentication, validation, database operations, and **exception handling** for common application workflows.
  - Implemented request logging and HTTP error handling to support **diagnosis of API, integration, and database issues**.
  - Containerized the application with Docker and documented common deployment and troubleshooting procedures.

Publications

- **Deep Learning Based Brain Tumor Classification using Orange Data Analytics** *2026*
  IEEE International Conference (SCEECS 2026) View Paper
  - Developed and evaluated a deep learning pipeline using a **16,000+ MRI image dataset** aggregated from Kaggle, BRISC, and FigShare.
  - Applied **data preprocessing, PCA, augmentation, and stratified 10-fold cross-validation**, reducing feature dimensionality by **75%** while retaining **95% variance**.
  - Achieved **96%+ accuracy** and **0.997 AUC**, demonstrating strong analytical evaluation across multiple performance metrics.

Achievements

- Selected as a **Software Engineering Fellow at Headstarter AI**, working on real-world software engineering and AI projects.
- Secured **3rd Place** in the **CodeMavens 2025 Hackathon** organized by the ACM-W SRM Student Chapter.
- Published research on **deep learning-based brain tumor classification** at the **IEEE International Conference (SCEECS 2026)**.

Technical Skills

- **Languages**: Java, Python, C++, JavaScript, TypeScript, SQL
- **Networking & Systems**: Computer Networks, TCP/IP, HTTP/HTTPS, DNS, Linux, Operating Systems
- **Software Engineering**: REST APIs, Spring Boot, FastAPI, API Design, Authentication, Authorization, Debugging
- **Databases**: MySQL, PostgreSQL, MongoDB, SQL, Data Modeling, Query Optimization, Indexing
- **Testing & DevOps**: Unit Testing, Integration Testing, Docker, AWS, CI/CD, Git, GitHub
- **Core CS**: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Troubleshooting
- **Security Fundamentals**: Authentication, Authorization, VPN Fundamentals, Firewall Fundamentals